



3433 - Mapping star formation and feedback in clumpy galaxies at redshift ~ 5

Cycle: 2, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(1) MS1358-ARC
	8		NIRSpec IFU Spectroscopy	(1) MS1358-ARC
	2		NIRSpec IFU Spectroscopy	(4) MACS0940-ARC
	3		NIRSpec IFU Spectroscopy	(4) MACS0940-ARC

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
	4		NIRSpec IFU Spectroscopy	(6) RCS0224-ARC
Observation Folder				
	5		NIRCam Imaging	(10) MS1358-IMAGING
	6		NIRCam Imaging	(11) MACS0940-IMAGING
	7		NIRCam Imaging	(12) RCS0224-IMAGING

ABSTRACT

We propose NIRSpec-IFU spectroscopy and NIRCam imaging to probe star formation and feedback in three exceptionally bright and extended, gravitationally lensed, high redshift ($z > 4$) galaxies, representative of the UV-selected population at these redshifts. Key emission lines (e.g. H α , H β , [OIII]) combined with existing MUSE/VLT rest-UV observations, are critical to map the gas physical conditions and kinematics from ~ 100 pc to kpc scales. NIRCam imaging (complemented with HST) will enable us to derive clump physical properties (size, ages, masses) down to 10s pc. We will:

1. probe disk fragmentation theories; 2. test predictions of increased clump star formation densities with redshift; 3. map the starburst propagation while investigating clump survival. Spatially resolving multiple nebular lines will map key parameters of the interstellar medium (ISM) and evaluate the role of clump feedback. We will trace the global kinematics of the ISM and investigate whether star-forming clumps show a trend in velocity dispersion. We will relate the distribution of ionizing radiation originating from the clumps with the global properties of each galaxy, and the overall morphology and kinematics of the circum-galactic medium seen in Ly α emission at large scales. These unprecedented studies are fundamental to uncover how star formation operates in normal galaxies at redshift beyond cosmic noon and inform increasingly detailed cosmological simulations. Extensive deep field campaigns will provide an incredible census of unresolved galaxies at these redshifts, but only studies as we propose here will enable us to dissect star formation and feedback that determine their evolution.

OBSERVING DESCRIPTION

We request NIRSpec IFU spectroscopy and NIRCam imaging of three lensed galaxies at $z > 4$ (MACS0940, RCS0224, MS1358) to measure the resolved spectral line and SED properties of star forming regions across each galaxy. The targets are selected to be extremely magnified ($\mu > 10$, straddling on the lensing critical line) by massive lensing clusters at $z = 0.3-0.7$. The NIRSpec observations are taken in G395H (2 integrations per exposure, 20 groups) for all targets, as well as G235M (2 integrations per exposure, 16 groups) for MACS0940. For both configurations the detector setup is NRSIRS2 and an 8 or 4 point medium cycling dithering pattern is used. The corresponding exposure time allows us to detect H α emission at $> 5\sigma$ (peak wavelengths, 0.2arcsec diameter aperture for a point source in the faintest clumps $H_{AB}=28$) while recovering additional lines (e.g. [OIII], H β , [NII], [SII]) at $> 3\sigma$ in multiple regions across each galaxy. We require a specific range of PA to optimise the orientation

JWST Proposal 3433 (Created: Tuesday, April 9, 2024 at 2:00:35 PM Eastern Standard Time) - Overview
of the IFU with the elongation of MS1358 and MACS0940.

NIRCam imaging is requested in 4 filters: F115W, F200W, F277W and F356W, using a MEDIUM8 readout pattern, 3 dither positions and 10 group per integration. The corresponding exposure times allows to reach $\text{SNR} > 5-10$ in all filters for the faintest detect clumps ($H_{\text{AB}}=28$). Each target is centered within a NIRCam detector for maximum exposure time, while the module covers the cluster core and additional background galaxies.

Proposal 3433 - Targets - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	MS1358-ARC	RA: 13 59 48.6834 (209.9528475d) Dec: +62 30 48.53 (62.51348d) Equinox: J2000	Epoch of Position: 2000.0	
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]				
	(4)	MACS0940-ARC	RA: 09 40 53.6792 (145.2236633d) Dec: +07 44 15.69 (7.73769d) Equinox: J2000	Epoch of Position: 2000.0	
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]				
	(6)	RCS0224-ARC	RA: 02 24 33.8332 (36.1409717d) Dec: -00 02 18.36 (-.03843d) Equinox: J2000	Epoch of Position: 2000.0	
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]				
	(10)	MS1358-IMAGING	RA: 13 59 50.3451 (209.9597712d) Dec: +62 30 52.67 (62.51463d) Equinox: J2000	Epoch of Position: 2000.0	
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]				
	(11)	MACS0940-IMAGING	RA: 09 40 53.9139 (145.2246412d) Dec: +07 44 25.44 (7.74040d) Equinox: J2000	Epoch of Position: 2000.0	
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]				
	(12)	RCS0224-IMAGING	RA: 02 24 34.1151 (36.1421463d) Dec: -00 02 32.08 (-.04224d) Equinox: J2000	Epoch of Position: 2000.0	
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]				

Proposal 3433 - Observation 1 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Observation	Proposal 3433, Observation 1											Tue Apr 09 19:00:35 GMT 2024	
	Diagnostic Status: Warning												
	Observing Template: NIRSspec IFU Spectroscopy												
Diagnostics	(Observation 1) Warning (Form): Verification imaging with an ALLOPEN MSA configuration may cause persistence from bright sources in the field of view.												
	(Observation 1) Warning (Form): WATA may be a better choice for more accurate placement in the IFU for point-like acquisition targets with positional uncertainty of 0.1 arcsec or less.												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	MS1358-ARC		RA: 13 59 48.6834 (209.9528475d)			Epoch of Position: 2000.0						
				Dec: +62 30 48.53 (62.51348d)									
				Equinox: J2000									
Template	Comments:												
	Category=Galaxy												
	Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]												
Template	TA Method												
	VERIFY_ONLY												
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			MEDIUM		1		8				
Pointing Verification	#	PV MSA Configuration		Filter	PV Readout Pattern		PV Groups/Int	PV Integrations/Exp		PV Total Dithers	PV Total Integrations	PV Total Exposure Time	
	1	ALLOPEN		F140X	NRSIRS2RAPID		8	1		1	1	131.3	
Spectral Elements	#	Grating/Filter		Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP		NRSIRS2	20	2	false	true	NONE	8	16	23575.646	

Proposal 3433 - Observation 1 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Special Requirements

Aperture PA Range 70 to 110 Degrees (V3 291.02835083 to 331.02835083)
Aperture PA Range 160 to 200 Degrees (V3 21.02835083 to 61.02835083)
Aperture PA Range 250 to 290 Degrees (V3 111.02835083 to 151.02835083)
Aperture PA Range 340 to 20 Degrees (V3 201.02835083 to 241.02835083)

Proposal 3433 - Observation 8 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Observation	Proposal 3433, Observation 8												Tue Apr 09 19:00:35 GMT 2024		
	Diagnostic Status: Warning														
Observing Template: NIRSspec IFU Spectroscopy															
Diagnostics	(Observation 8) Warning (Form): Verification imaging with an ALLOPEN MSA configuration may cause persistence from bright sources in the field of view.														
	(Observation 8) Warning (Form): WATA may be a better choice for more accurate placement in the IFU for point-like acquisition targets with positional uncertainty of 0.1 arcsec or less.														
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.														
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous					
	(1)	MS1358-ARC		RA: 13 59 48.6834 (209.9528475d)			Epoch of Position: 2000.0								
				Dec: +62 30 48.53 (62.51348d)											
				Equinox: J2000											
Template	Comments:														
	Category=Galaxy														
	Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]														
Template	TA Method														
	VERIFY_ONLY														
Dithers	#	Dither Type			Size		Starting Point		Number of Points			Points			
	1	CYCLING			MEDIUM		1		8						
Pointing Verification	#	PV MSA Configuration		Filter	PV Readout Pattern		PV Groups/Int	PV Integrations/Exp		PV Total Dithers	PV Total Integrations		PV Total Exposure Time		
	1	ALLOPEN		F140X	NRSIRS2RAPID		8	1		1	1		131.3		
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID			
	1	G395H/F290LP	NRSIRS2	20	2	false	true	NONE	8	16	23575.646				

Proposal 3433 - Observation 8 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Special Requirements	Aperture PA Range 70 to 110 Degrees (V3 291.02835083 to 331.02835083) Aperture PA Range 160 to 200 Degrees (V3 21.02835083 to 61.02835083) Aperture PA Range 250 to 290 Degrees (V3 111.02835083 to 151.02835083) Aperture PA Range 340 to 20 Degrees (V3 201.02835083 to 241.02835083)
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Proposal 3433 - Observation 2 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Observation	Proposal 3433, Observation 2											Tue Apr 09 19:00:35 GMT 2024
	Diagnostic Status: Warning											
Observing Template: NIRSspec IFU Spectroscopy												
Diagnostics	(Observation 2) Warning (Form): Verification imaging with an ALLOPEN MSA configuration may cause persistence from bright sources in the field of view.											
	(Observation 2) Warning (Form): WATA may be a better choice for more accurate placement in the IFU for point-like acquisition targets with positional uncertainty of 0.1 arcsec or less.											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(4)	MACS0940-ARC	RA: 09 40 53.6792 (145.2236633d)			Epoch of Position: 2000.0						
			Dec: +07 44 15.69 (7.73769d)									
			Equinox: J2000									
Template	Comments:											
	Category=Galaxy											
Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]												
Template	TA Method											
	VERIFY_ONLY											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		MEDIUM		1		8				
Pointing Verification	#	PV MSA Configuration	Filter	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Dithers	PV Total Integrations	PV Total Exposure Time			
	1	ALLOPEN	F140X	NRSIRS2RAPID	8	1	1	1	131.3			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2	20	2	false	true	NONE	8	16	23575.646	

Proposal 3433 - Observation 2 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Special Requirements

Aperture PA Range 25 to 65 Degrees (V3 246.02835083 to 286.02835083)
Aperture PA Range 115 to 155 Degrees (V3 336.02835083 to 16.02835083)
Aperture PA Range 205 to 245 Degrees (V3 66.02835083 to 106.02835083)
Aperture PA Range 295 to 335 Degrees (V3 156.02835083 to 196.02835083)

Same Aperture PA 2, 3

Proposal 3433 - Observation 3 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Observation	Proposal 3433, Observation 3												Tue Apr 09 19:00:35 GMT 2024
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Observation 3) Warning (Form): Verification imaging with an ALLOPEN MSA configuration may cause persistence from bright sources in the field of view.												
	(Observation 3) Warning (Form): WATA may be a better choice for more accurate placement in the IFU for point-like acquisition targets with positional uncertainty of 0.1 arcsec or less.												
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(4)	MACS0940-ARC		RA: 09 40 53.6792 (145.2236633d)			Epoch of Position: 2000.0						
				Dec: +07 44 15.69 (7.73769d)									
				Equinox: J2000									
Template	Comments:												
	Category=Galaxy												
	Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]												
TA Method	TA Method												
	VERIFY_ONLY												
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			MEDIUM		1		4				
Pointing Verification	#	PV MSA Configuration		Filter	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Dithers	PV Total Integrations	PV Total Exposure Time			
	1	ALLOPEN		F140X	NRSIRS2RAPID	8	1	1	1	131.3			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G235M/F170LP	NRSIRS2	16	2	false	true	NONE	4	8	9453.601		

Proposal 3433 - Observation 3 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Special Requirements	Same Aperture PA 2, 3
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Proposal 3433 - Observation 4 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Observation	Proposal 3433, Observation 4											Tue Apr 09 19:00:35 GMT 2024
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Observation 4) Warning (Form): Verification imaging with an ALLOPEN MSA configuration may cause persistence from bright sources in the field of view.											
	(Observation 4) Warning (Form): WATA may be a better choice for more accurate placement in the IFU for point-like acquisition targets with positional uncertainty of 0.1 arcsec or less.											
(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(6)	RCS0224-ARC	RA: 02 24 33.8332 (36.1409717d) Dec: -00 02 18.36 (-.03843d) Equinox: J2000			Epoch of Position: 2000.0						
Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]												
Template	TA Method											
	VERIFY_ONLY											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		MEDIUM		1		8				
Pointing Verification	#	PV MSA Configuration	Filter	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Dithers	PV Total Integrations	PV Total Exposure Time			
	1	ALLOPEN	F140X	NRSIRS2RAPID	8	1	1	1	131.3			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2	20	2	false	true	NONE	8	16	23575.646	

Proposal 3433 - Observation 5 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Observation	Proposal 3433, Observation 5										Tue Apr 09 19:00:35 GMT 2024								
	Diagnostic Status: Warning																		
	Observing Template: NIRCam Imaging																		
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.																		
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous									
	(10)	MS1358-IMAGING	RA: 13 59 50.3451 (209.9597712d)				Epoch of Position: 2000.0												
			Dec: +62 30 52.67 (62.51463d)																
			Equinox: J2000																
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]																		
Template	Module				Subarray				Target Placement										
	ALL				FULL				Module Gap										
Dithers	#	Primary Dither Type			Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions								
	1	INTRAMODULE			3		STANDARD				1								
Spectral Elements	#	Short Filter		Long Filter		Readout Pattern		Groups/Int		Integrations/Exp		Total Integrations		Total Dithers		Total Exposure Time		ETC Wkbk.Calc ID	
	1	F115W		F277W		MEDIUM8		10		1		3		3		3156.61			
	2	F200W		F444W		MEDIUM8		10		1		3		3		3156.61			
Special Requirements	Aperture PA Range 81.92542306 to 93.92542306 Degrees (V3 82.0 to 94.0) Aperture PA Range 101.92542306 to 107.92542306 Degrees (V3 102.0 to 108.0) Aperture PA Range 110.92542306 to 170.92542306 Degrees (V3 111.0 to 171.0) Aperture PA Range 181.92542306 to 185.92542306 Degrees (V3 182.0 to 186.0) Aperture PA Range 187.92542306 to 188.92542306 Degrees (V3 188.0 to 189.0) Aperture PA Range 195.92542306 to 226.92542306 Degrees (V3 196.0 to 227.0) Aperture PA Range 227.92542306 to 243.92542306 Degrees (V3 228.0 to 244.0) Aperture PA Range 250.92542306 to 331.92542306 Degrees (V3 251.0 to 332.0) Aperture PA Range 340.92542306 to 348.92542306 Degrees (V3 341.0 to 349.0) Aperture PA Range 355.92542306 to 76.92542306 Degrees (V3 356.0 to 77.0) Offset 45.0 arcsec, -26.0 arcsec																		

Proposal 3433 - Observation 6 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Observation	Proposal 3433, Observation 6										Tue Apr 09 19:00:35 GMT 2024
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRCcam Imaging										
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(11)	MACS0940-IMAGING	RA: 09 40 53.9139 (145.2246412d) Dec: +07 44 25.44 (7.74040d) Equinox: J2000			Epoch of Position: 2000.0					
Template	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]										
	Module		Subarray				Target Placement				
Dithers	ALL		FULL				Module Gap				
	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
Spectral Elements	1	INTRAMODULE		3		STANDARD				1	
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
Special Requirements	1	F115W	F277W	MEDIUM8	10	1	3	3	3156.61		
	2	F200W	F444W	MEDIUM8	10	1	3	3	3156.61		
	Aperture PA Range 64.92542306 to 293.92542306 Degrees (V3 65.0 to 294.0) Aperture PA Range 299.92542306 to 303.92542306 Degrees (V3 300.0 to 304.0) Aperture PA Range 305.92542306 to 60.92542306 Degrees (V3 306.0 to 61.0) Offset 50.0 arcsec, -31.0 arcsec										

Proposal 3433 - Observation 7 - Mapping star formation and feedback in clumpy galaxies at redshift ~5

Observation	Proposal 3433, Observation 7										Tue Apr 09 19:00:35 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(12)	RCS0224-IMAGING	RA: 02 24 34.1151 (36.1421463d)			Epoch of Position: 2000.0					
			Dec: -00 02 32.08 (-.04224d)								
			Equinox: J2000								
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies]										
Template	Module			Subarray			Target Placement				
	ALL			FULL			Module Gap				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULE		3		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F115W	F277W	MEDIUM8	10	1	3	3	3156.61		
	2	F200W	F444W	MEDIUM8	10	1	3	3	3156.61		
Special Requirements	Aperture PA Range 65.42542306 to 143.92542306 Degrees (V3 65.5 to 144.0) Aperture PA Range 151.92542306 to 199.92542306 Degrees (V3 152.0 to 200.0) Aperture PA Range 213.92542306 to 214.92542306 Degrees (V3 214.0 to 215.0) Aperture PA Range 227.92542306 to 229.92542306 Degrees (V3 228.0 to 230.0) Aperture PA Range 234.92542306 to 49.92542306 Degrees (V3 235.0 to 50.0) Offset 49.0 arcsec, 37.0 arcsec										